

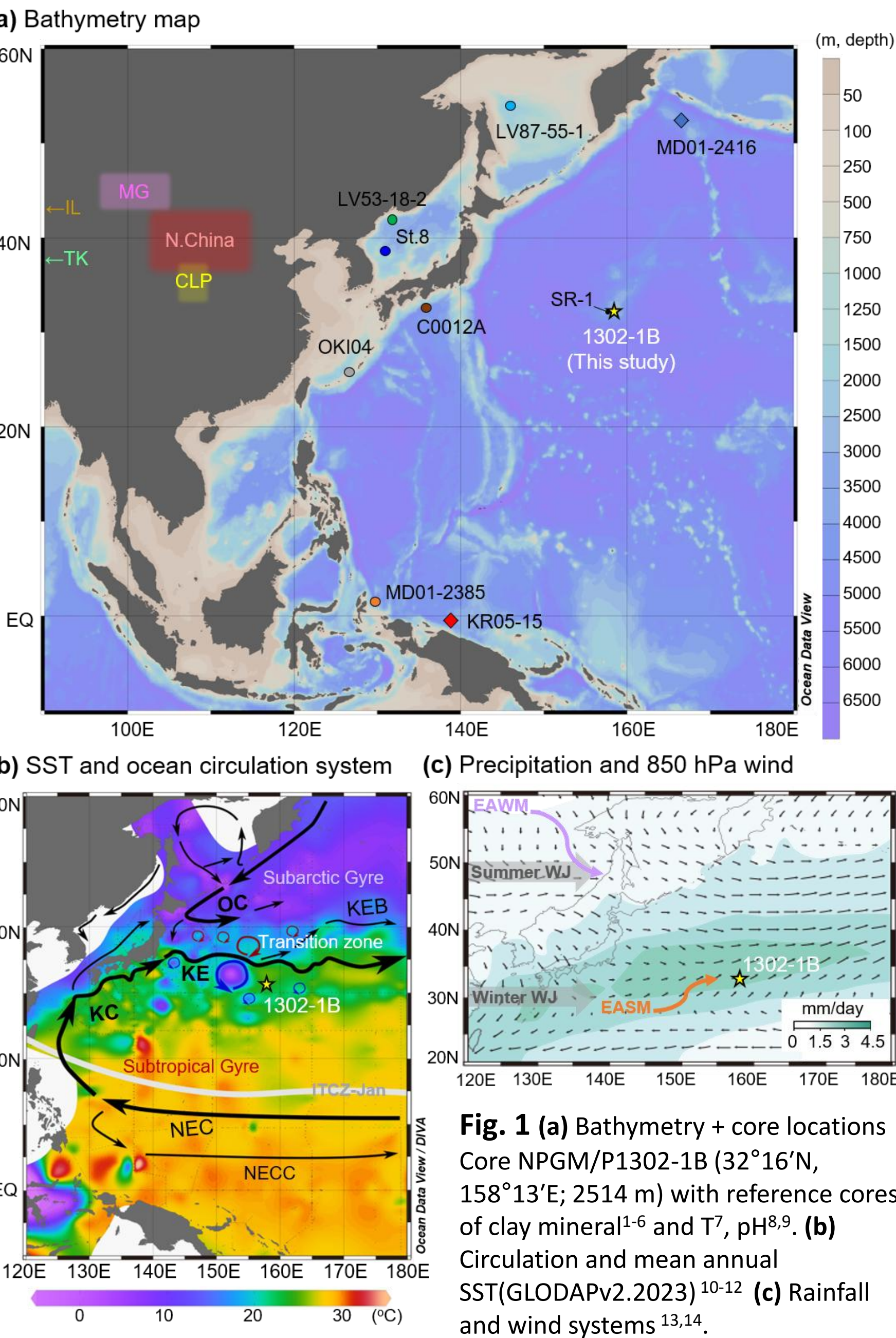
Linking East Asian Dust Flux to Surface Ocean Carbonate Chemistry in the Kuroshio-Oyashio Extension: A 23-kyr Record of Clay Mineralogy and Boron Isotopes

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Background & Aim

- NW.Pacific, KE (Kuroshio-oyashio Extension) area: subtropical/subpolar gyre boundary
- In this study:
- Atmospheric changes: Eolian provenance via clay mineral composition + magnetic properties
- Oceanographic changes: Multi-depth pH from foraminiferal $\delta^{11}\text{B}$ (3 species)
- Understanding the atmospheric + oceanographic forcing during the last deglaciation (~23 ka)



Foraminifera species thriving depth
@ Mixed layer: *G. bulloides* (~50 m), *N. pachyderma* (~20-75 m)
@ Below DCM: *T. sacculifer* with sac (> 75 m)
The DCM is estimated at ~50–75 m depth in study area

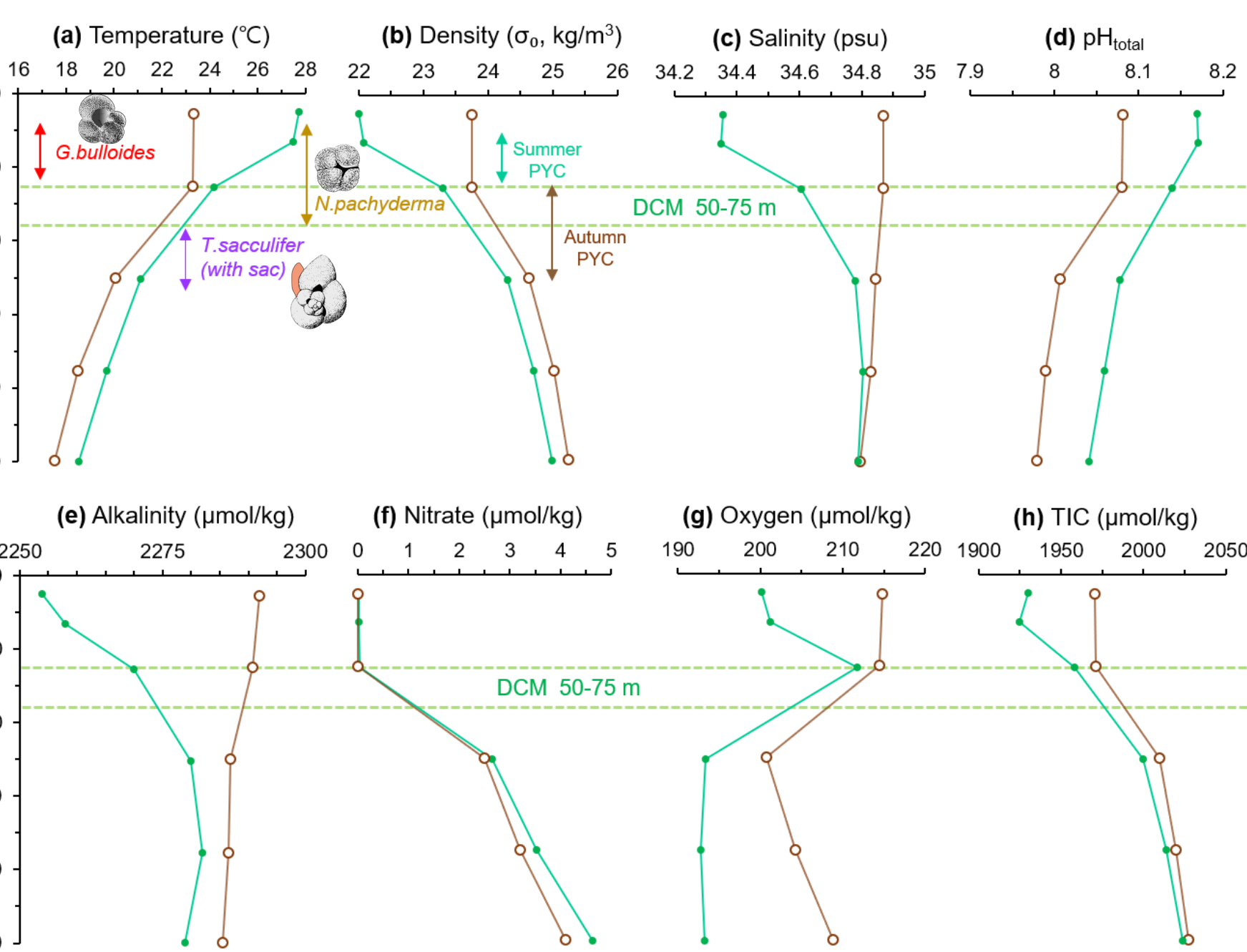


Fig. 2 Depth profiles of seawater chemistry (Aug 1999¹⁵, Nov 2004¹⁶) from the NWP (CTD stations data @ 32°01'N, 155°04'E; GLODAPv2.2023).
¹⁵R/V Hakurei Maru-II cruise NH99, ¹⁶R/V Mirai cruise MR04-07

Methods

- Age model**
•¹⁴C AMS dating of mixed planktic foraminifera (Beta Analytic, USA), Cal BP using Calib v7.1; error < 70 yr, reservoir age 414 yr¹⁷
- Boron isotope & trace element analysis**
• $\delta^{11}\text{B}$ of *G. bulloides* (300–355 μm), *N. pachyderma* (250–300 μm), and *T. sacculifer* with sac (425–800 μm) (MC-ICP-MS: NuPlasma3, SNU IRF) after column separation¹⁸; trace elements (Al, Ca, Mn, Sr) by ICP-MS (Elan DRC-II, KBSI)
- Clay mineral analysis**
•XRD of <2 μm oriented samples (air-dried & ethylene glycol-saturated); Bruker D8 Advance A25 (GNU), Cu K α ($\lambda = 1.541 \text{ \AA}$); obtain Illite crystallinity (IC)–FWHM of illite 10 $\text{\AA}$ peak, Illite chemical index (ICI) –5 $\text{\AA}$ to 10 $\text{\AA}$ peak area ratio and proportions of illite, smectite, kaolinite, chlorite by weighting factors¹⁹
- Grain-size end-member modelling (EMMA)**
•Laser diffraction particle size analysis (Beckman Coulter LS 13 320 XR); 5-end-member model fitted using AnalySize v1.1²⁰

Depth-dependent pH divergence peaks at LGM termination

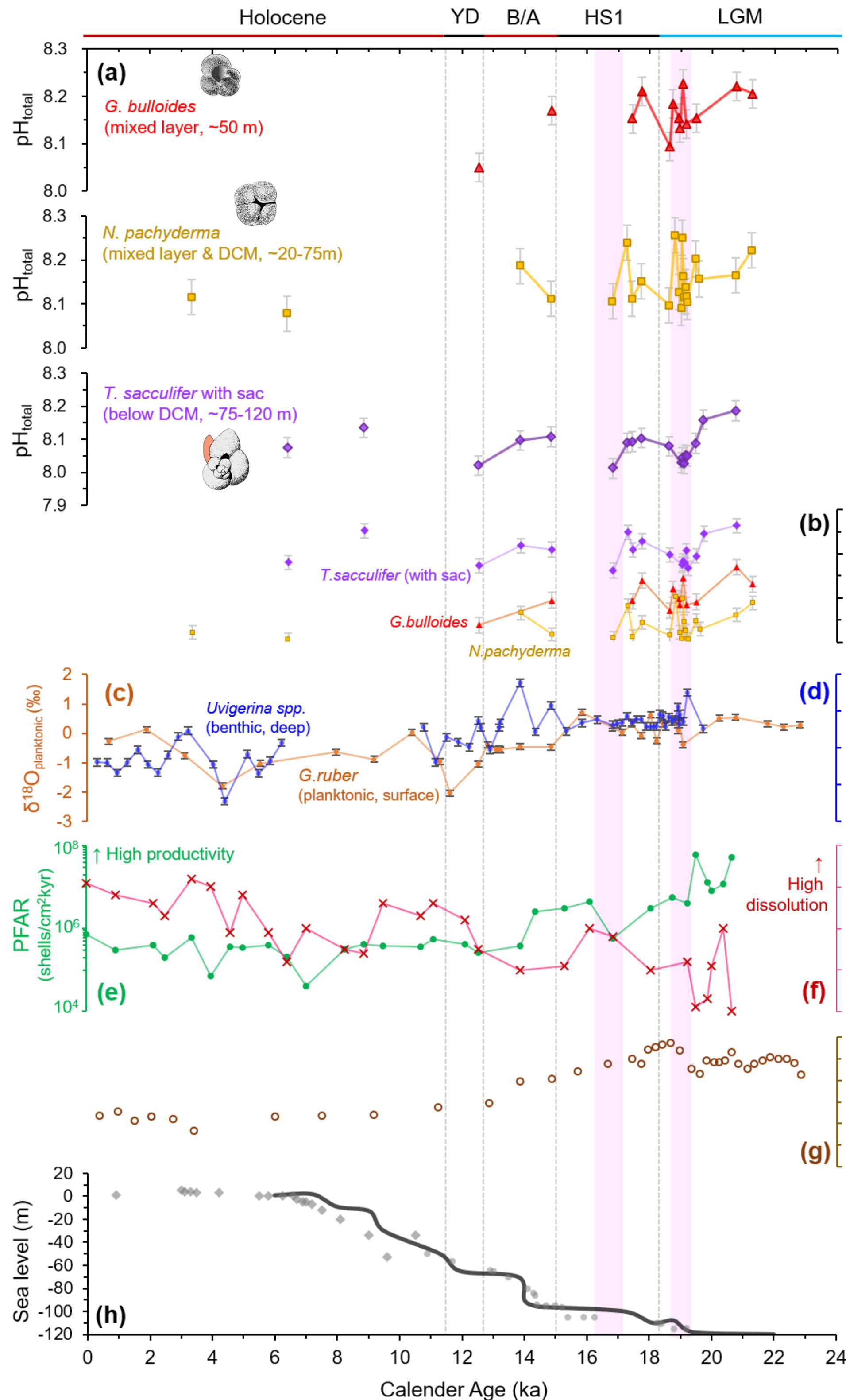
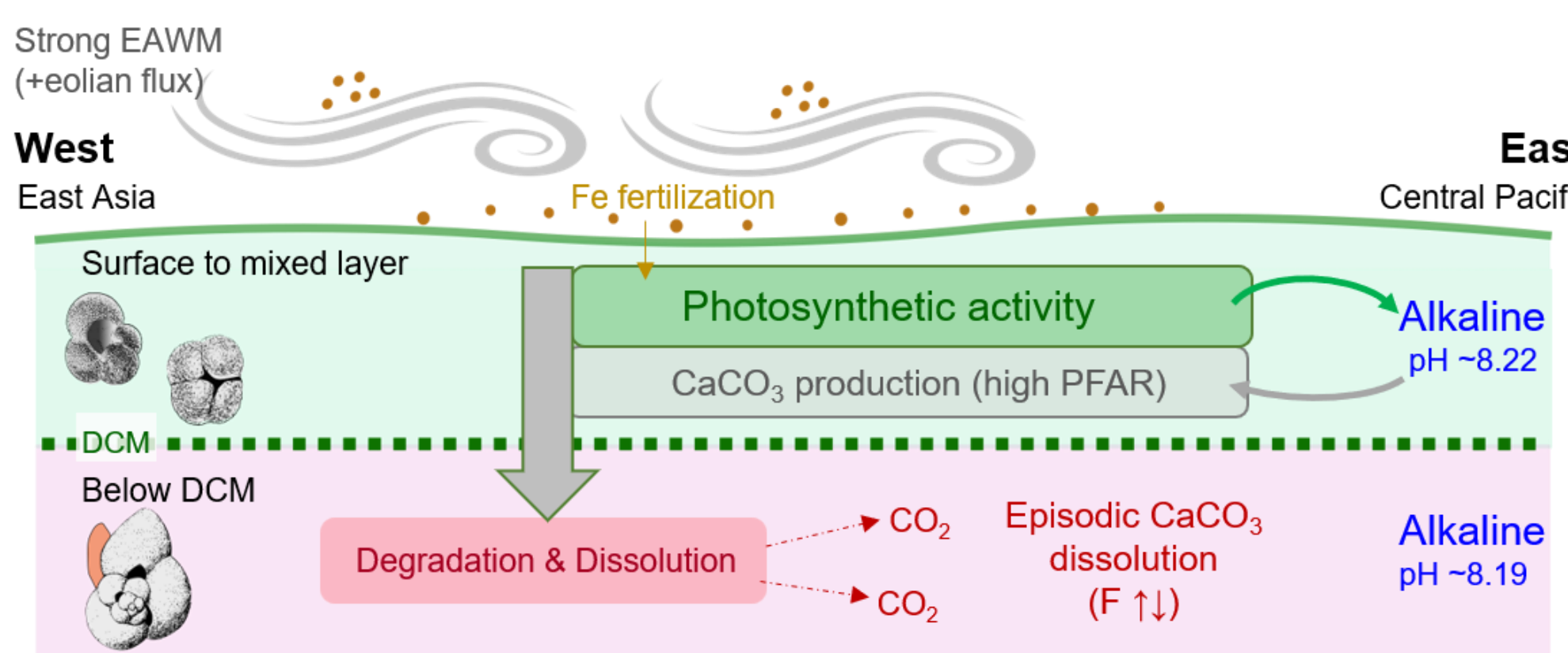
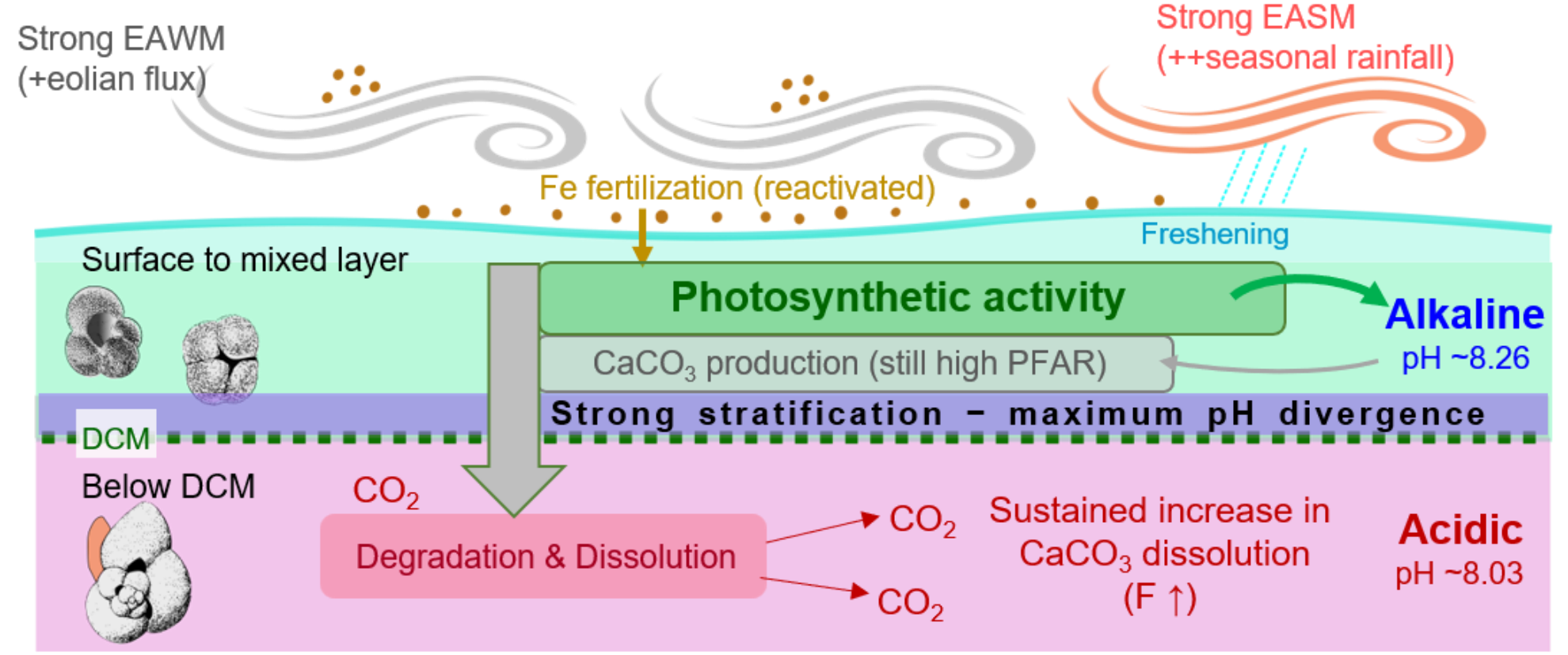


Fig. 3 (a) Boron isotope-derived pH_{total} (b) $\delta^{11}\text{B}$ of planktic foraminiferal species (c) Planktic and (d) benthic foraminiferal $\delta^{18}\text{O}$ (e) Planktic foraminiferal accumulation rate (PFAR) and (f) fragment factor (F%) for carbonate dissolution²¹. (g) Eolian component (wt%)²². (h) Sea level of NWP^{23–25}. Pink=rapid acidification events

(a) Late LGM (~21.3–19.4 ka)



(b) LGM termination (19.3–18.7 ka)



(c) Mid-HS1 (~16.8 ka)

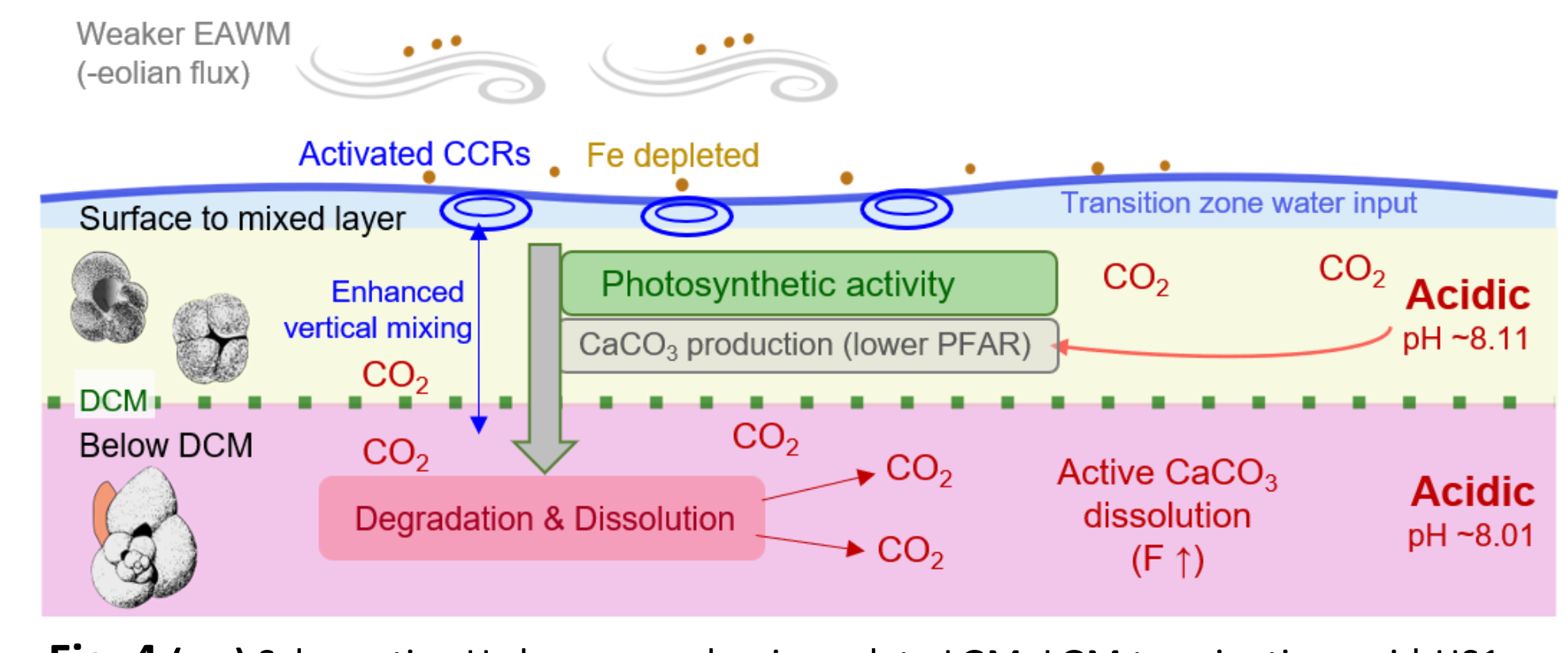


Fig. 4 (a-c) Schematic pH change mechanisms: late LGM, LGM termination, mid-HS1. Arrow thickness denotes relative intensity of eolian flux and organic matter export.

Strong EASM + EAWM → High eolian flux + seasonal rainfall
→ Fe fertilization by dust + stratification in seawater
→ depth-dependent pH divergence at LGM termination

Eolian surge ← monsoon co-intensification, exposed continental shelves

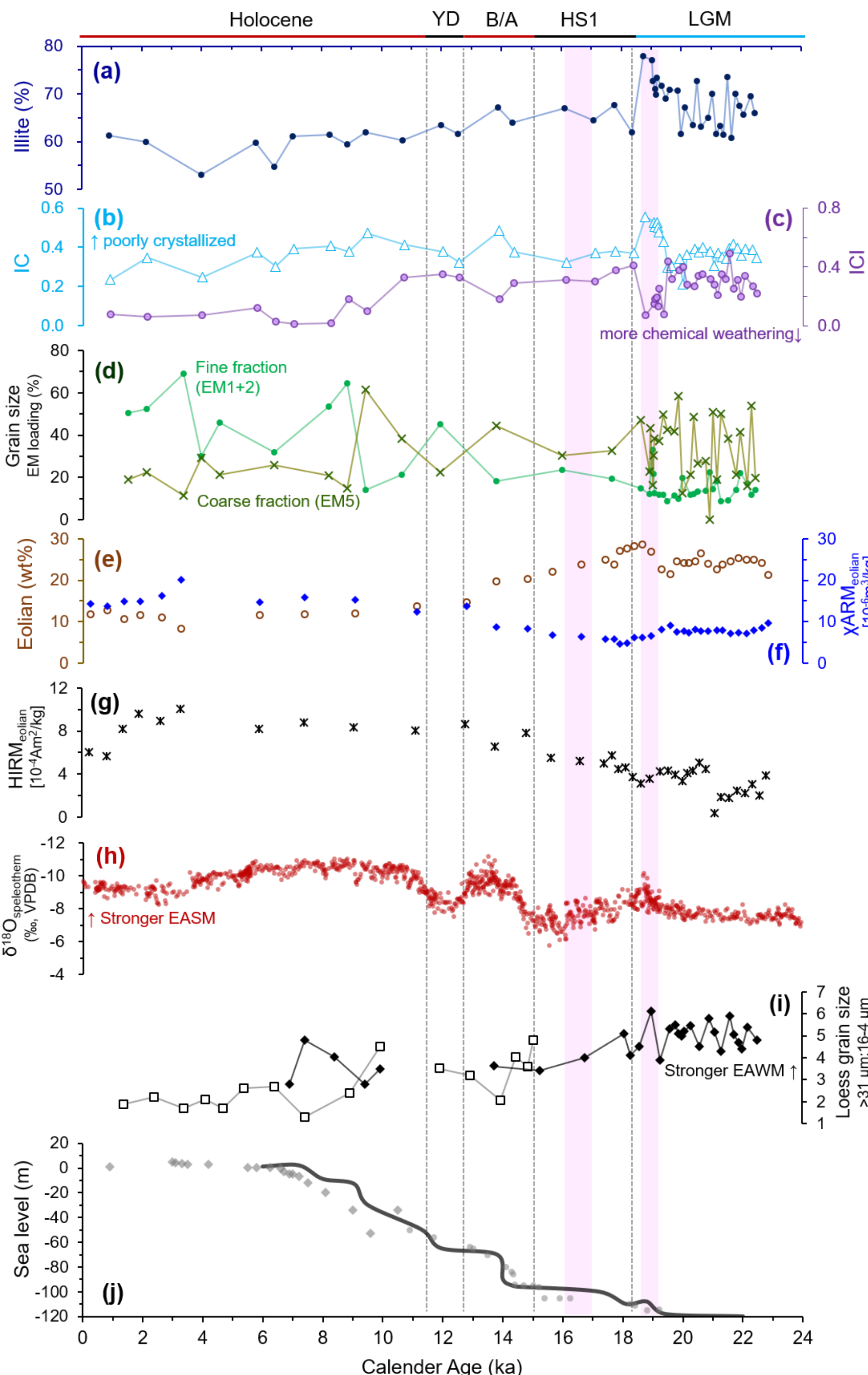


Fig. 5 (a) Illite (%), (b) IC (Illite crystallinity), (c) ICI (Illite chemical index), (d) Grain size end-member loadings (EM1+2 <2 μm ; EM5 ~35 μm), (e) Eolian (wt%)²², (f) $\chi_{\text{ARM}}^{\text{eolian}}$ ²² (fine-grained magnetite in eolian fraction), (g) $\text{HIRM}_{\text{eolian}}$ ²² (hematite/goethite in eolian fraction), (h) Speleothem $\delta^{18}\text{O}$ (EASM intensity)^{24,25}. (i) Loess grain size (EAWM intensity)²⁸, (j) Sea level of NWP^{23–25}.

High illite % = dominant Asian dust provenance

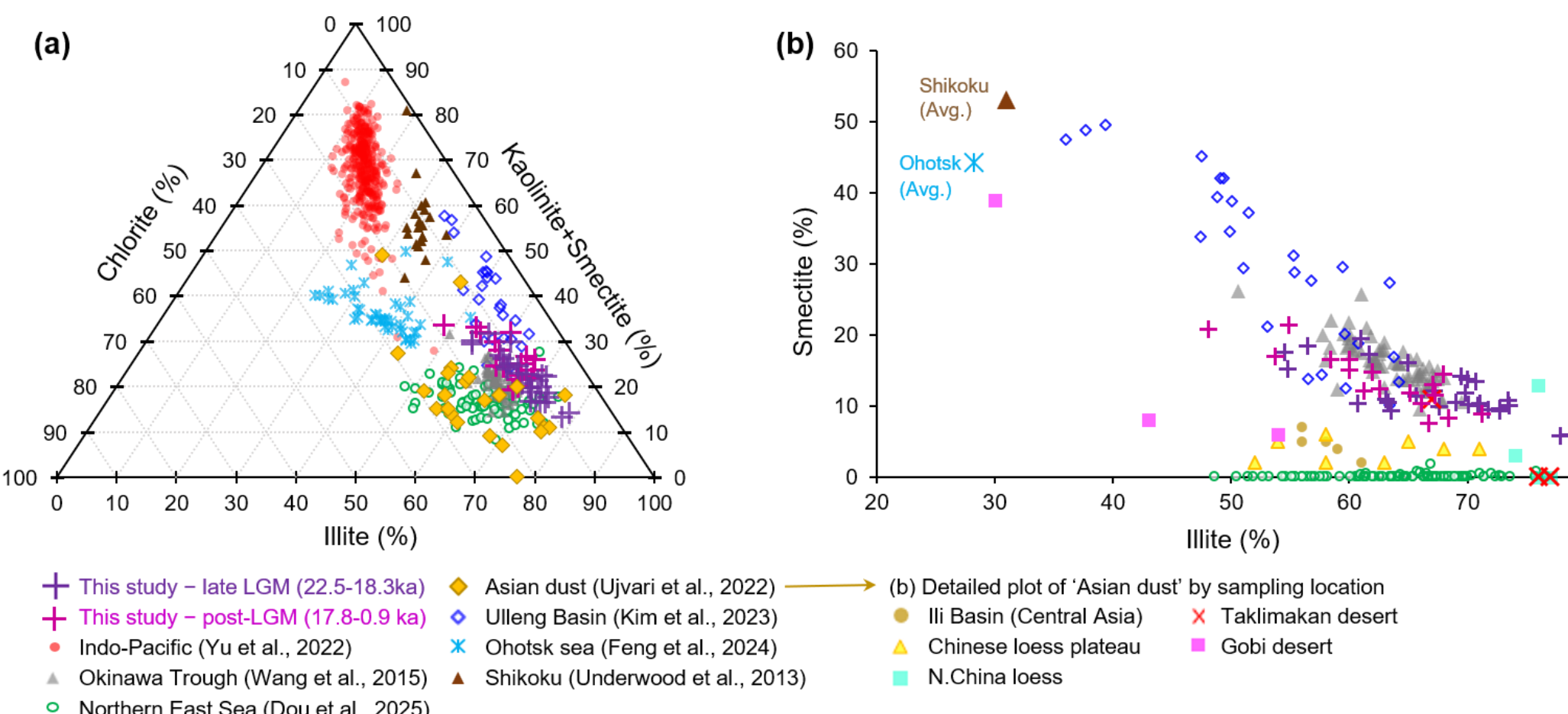


Fig. 6 Clay mineral composition of the study core and potential source regions^{1–6,29}
(a) Ternary diagram of smectite-illite-(chlorite+kaolinite), (b) Cross plot of illite vs smectite

pH divergence across western Pacific

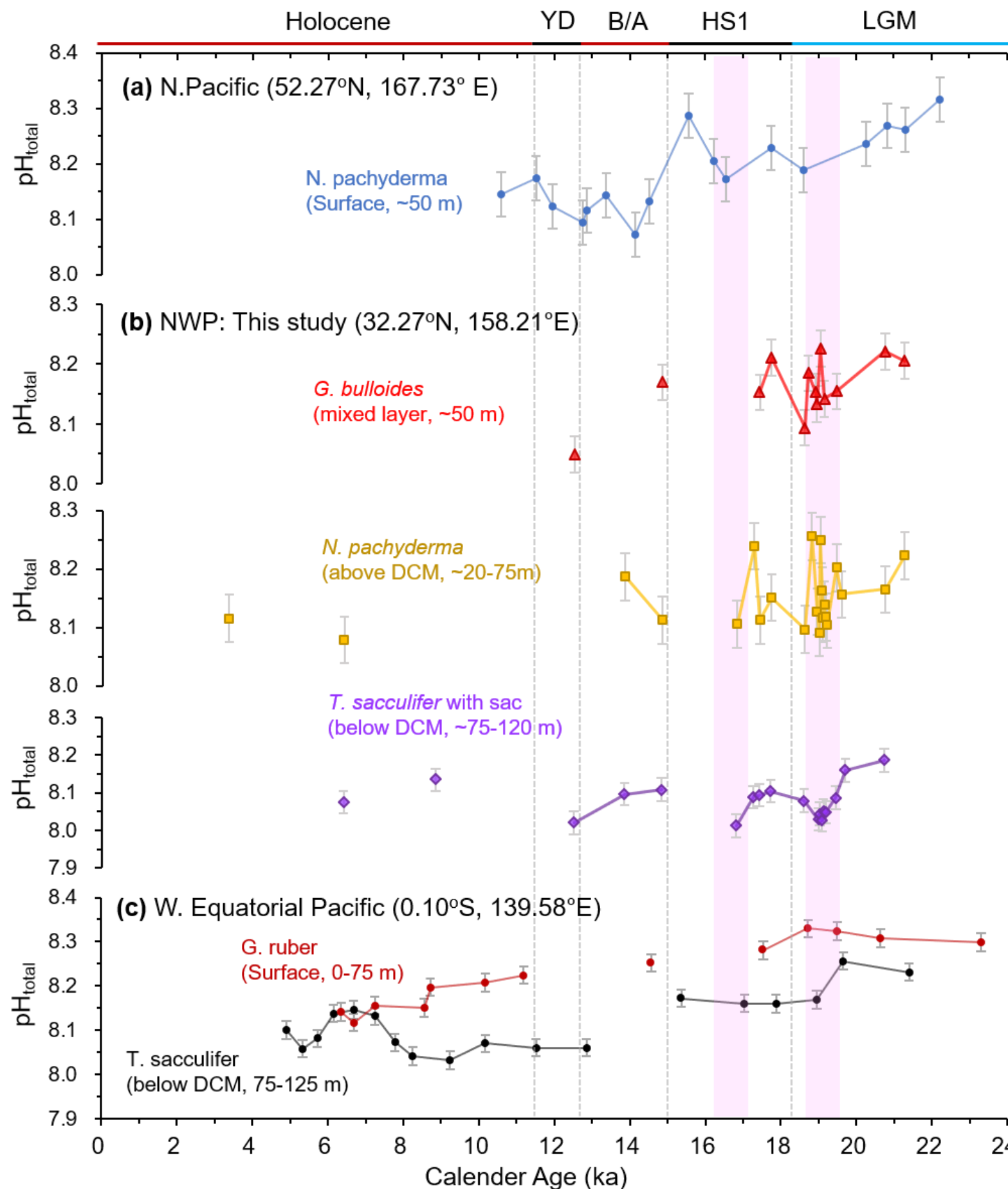


Fig. 7 pH records from (a) N. Pacific⁸, (b) NWP (this study), (c) W. equatorial Pacific⁹